AC4409

Corporate Financing

Winter 2024 Examination

Paper 1 · 1.5 hours · Answer ALL questions · Show ALL workings

Module	AC4409 — Corporate Financing
Session	Winter 2024
Structure	Two questions, unequal weight — Question 1 worth 63 marks, Question 2 worth 37 marks. Both compulsory (unlike other AC4409 papers you may have seen, which offer a choice of questions).
This scheme covers	Both questions in full, including the "pick the correct statements" parts in each question.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the method marks for those later steps. This applies throughout the venture capital and MM leverage workings in Question 1.

On the "pick the correct statements" parts (Q1(c) and Q2(c)). Both papers state that picking any incorrect statement zeroes the mark for that part, with partial credit for identifying some but not all of the correct ones. This scheme identifies exactly which statements are correct/false and explains why, so you can see the reasoning behind each — but under the exam's own stated rule, no explanation is required or rewarded in your actual answer; only the selection itself is marked.

Question 1

Venture Financing & Modigliani-Miller (63 Marks)

Part (a)(i)

15 Marks

Starware Software was founded last year to develop software for gaming applications. The founder initially invested €900,000 and received 11 million shares of stock. Starware now needs to raise a second round of capital, and it has identified an interested venture capitalist. This venture capitalist will invest €1 million and wants to own 37% of the company after the investment is completed.

(i) How many shares must the venture capitalist receive to end up with 37% of the company? What is the implied price per share of this funding round?

FULL MODEL ANSWER

The venture capitalist's 37% stake is a share of the **post-money** company (existing shares plus the new shares issued to the VC) — not 37% of the existing 11 million shares. Letting *x* be the number of new shares issued:

x ÷ (11,000,000 + x) = 0.37
x = 0.37 × (11,000,000 + x)
x × (1 − 0.37) = 0.37 × 11,000,000
x = 4,070,000 ÷ 0.63
x ≈ 6,460,317 new shares

Implied price per share = €1,000,000 ÷ 6,460,317
≈ €0.1548 per share

A very common error is computing 37% of the **existing** 11 million shares (giving 4,070,000 shares) rather than 37% of the **post-money** total — that approach ignores that the new shares themselves are part of the base the 37% is measured against, and understates the shares actually needed. The correct setup treats the new shares as part of both the numerator and the denominator.

How the 15 Marks Are Likely Allocated

Tier	Marks	What separates this tier
12–15	12–15	Correctly sets up the post-money ownership equation (new shares over existing plus new shares), solves for approximately 6,460,317 new shares, and correctly divides the €1m investment by that figure to reach the implied price per share.
7–11	7–11	Correct concept but computes 37% of the existing 11 million shares instead of the post-money total, or an arithmetic slip in solving the equation.
0–6	0–6	No attempt, or no understanding of how ownership percentage relates to shares issued in a financing round.

Part (a)(ii)

8 Marks

(ii) What will the value of the whole firm be after this investment (the post-money valuation)?

FULL MODEL ANSWER

The quickest route is the standard venture financing shortcut: post-money valuation = the new investment divided by the percentage of the company it buys. This works precisely because the VC's price per share was set (in part (i)) to make this true by construction:

Post-money valuation = Investment ÷ Ownership %
= €1,000,000 ÷ 0.37
≈ €2,702,703

This can equally be cross-checked by multiplying the implied price per share from part (i) by the *total* post-money share count (17,460,317 shares): €0.1548 × 17,460,317 ≈ €2,702,703 — the same answer either way, which is a useful check if the two parts are attempted out of order.

How the 8 Marks Are Likely Allocated

Tier	Marks	What separates this tier
6–8	6–8	Correctly divides the €1m investment by the 37% stake to reach ≈€2,702,703 (or cross-checks via price × total shares to the same figure).
3–5	3–5	Correct concept but an arithmetic slip, or the pre-money value used instead of post-money.
0–2	0–2	No attempt, or confuses post-money valuation with the investment amount itself.

You are an entrepreneur starting a biotechnology firm. If your research is successful, the technology can be sold for €35 million. If your research is unsuccessful, it will be worth nothing. To fund your research, you need to raise €2.7 million today. Investors are willing to provide you with €2.7 million in initial capital in exchange for 25% of the unlevered equity in the firm. Suppose this is a world of perfect capital markets:

Part (b)(i) **6 Marks**

(i) What is the total market value of the firm without leverage today?

FULL MODEL ANSWER

Investors paid €2.7 million for a 25% stake — grossing that up to the whole company gives the total unlevered firm value:

Unlevered firm value = Investment ÷ % of equity acquired
= €2,700,000 ÷ 25%
= €10,800,000

The €35 million potential success payoff is **not** used directly in this calculation — the market has already priced the probability-weighted expected value of that uncertain outcome into the price investors were willing to pay for 25% of the equity. €10.8m *is* that market-implied value; there's no need (or way, without an explicit probability of success) to derive it independently from the €35m figure.

How the 6 Marks Are Likely Allocated

Tier	Marks	What separates this tier
5–6	5–6	Correctly divides the €2.7m investment by the 25% stake to reach €10.8m, without attempting to independently derive a value from the €35m success payoff.
2–4	2–4	Correct concept but an arithmetic error, or unnecessary (and incorrect) use of the €35m figure.
0–1	0–1	No attempt, or values the firm using only the €35m success scenario.

Part (b)(ii) **14 Marks**

(ii) Suppose you borrow €0.9 million. According to the Modigliani and Miller (or simply MM) Proposition, what fraction of the firm’s equity must you sell to raise the additional €1.8 million you need?

FULL MODEL ANSWER

The whole point of the MM Proposition (in a world of perfect capital markets, with no taxes, transaction costs, or bankruptcy costs) is that a firm's **total** value doesn't depend on how it's financed — so borrowing €0.9 million doesn't change the total firm value found in part (i); it just changes how that value is split between debt and equity holders.

Step 1 — Value of the levered equity

	€
Total firm value (unchanged by MM, from part (i))	10,800,000
Less: new debt raised	(900,000)
Value of levered equity	**9,900,000**

Step 2 — How much equity still needs to be sold

	€
Total still required for the research	2,700,000
Less: amount raised via debt	(900,000)
Remaining amount to raise via equity	**1,800,000**

Step 3 — Fraction of equity to sell

Fraction of equity sold = €1,800,000 ÷ €9,900,000 (value of levered equity)
≈ 18.18%

A frequent error is dividing the €1.8m still needed by the **unlevered** equity value of €10.8m, rather than the **levered** equity value of €9.9m — but once €0.9m of debt is raised, the equity being sold is a claim on the remaining €9.9m of value, not the original €10.8m unlevered total.

How the 14 Marks Are Likely Allocated

Tier	Marks	What separates this tier
11–14	11–14	Correctly recognises total firm value is unchanged under MM, correctly derives the €9.9m levered equity value, correctly identifies the remaining €1.8m still needed, and divides to reach ≈18.18%.
6–10	6–10	Correct MM concept but divides by the wrong equity base (e.g. the €10.8m unlevered value), or an error in identifying the remaining amount needed.
0–5	0–5	No attempt, or no application of MM irrelevance at all (e.g. assumes total firm value changes because debt was raised).

Part (b)(iii) **10 Marks**

(iii) What is the value of your share of the firm’s equity in cases (i) and (ii)?

FULL MODEL ANSWER

This is the payoff of the whole exercise — MM's central insight, applied to the entrepreneur's own personal stake rather than just the firm's total value.

Case (i) — No leverage

Entrepreneur retains (1 − 25%) = 75% of the unlevered equity
Value = 75% × €10,800,000
= €8,100,000

Case (ii) — With €0.9m of leverage

Entrepreneur retains (1 − 18.18%) = 81.82% of the levered equity
Value = 81.82% × €9,900,000
= €8,100,000

The two figures are identical — exactly €8,100,000 either way. This is not a coincidence; it's MM's core message in miniature. Adding leverage doesn't create or destroy value for the entrepreneur under perfect capital markets: it simply changes how the fixed total pie is sliced between different classes of investors (debt vs. equity), while the entrepreneur's own retained slice, correctly valued, is unchanged. If your two answers here don't match, that's a strong signal to check the earlier parts for an arithmetic error rather than a sign that MM has somehow failed.

How the 10 Marks Are Likely Allocated

Tier	Marks	What separates this tier
8–10	8–10	Correctly calculates the entrepreneur's retained value in both cases, arrives at €8,100,000 in each, and explicitly notes that this equality demonstrates MM irrelevance for the entrepreneur's own stake.
4–7	4–7	Correct calculation in one case but an error (often carried from part (ii)) in the other, so the two figures don't match — the own-figure rule applies if the method is right but an earlier number was wrong.
0–3	0–3	No attempt, or no attempt to connect the two cases to the MM irrelevance result.

Part (c) **10 Marks**

Pick out all the correct statements from below (no need to explain; marks to be null for picking out any incorrect statement, and partly awarded for not picking out all the correct ones):

- The presence of financial distress costs can explain why firms choose debt levels that are too low to exploit the interest tax shield.
- The Tradeoff Theory suggests that a firm should choose a debt level where the tax savings from increasing leverage are just offset by the increased probability of incurring the costs of financial distress.
- Firms in industries such as real estate tend to have high distress costs because of a large proportion of tangible assets.
- Differences in the magnitude of financial distress costs and volatility of cash flows across industries do not impact the choice of leverage.
- The probability of financial distress depends on the likelihood that a firm will be unable to meet its debt commitments.
- As the level of debt increases the tax benefits of debt increase until interest costs exceed dividend payments.

FULL MODEL ANSWER

Correct statements: i, ii, and v.

i. TRUE. This is exactly the trade-off theory's explanation for why observed leverage is lower than a pure tax-shield-maximising strategy would suggest: if financial distress carried no cost, firms would gear up toward 100% debt to maximise the interest tax shield. The fact that they don't is explained by distress costs pulling the other way.

ii. TRUE. This is the trade-off theory's definition, almost verbatim: the optimal debt level is where the marginal tax benefit of one more euro of debt is exactly offset by the marginal expected cost of financial distress that additional debt creates.

iii. FALSE. This has the relationship backwards. Real estate is a standard example of an industry with **low** distress costs, precisely *because* its assets are tangible and easily resold or repurposed — a lender or acquirer can recover most of an asset's value even in a distressed sale. That's exactly why real estate firms are able to sustain *higher*, not lower, leverage than industries with intangible, firm-specific, or hard-to-redeploy assets (like technology or pharmaceuticals).

iv. FALSE. Differences in distress costs and cash flow volatility across industries are a central reason observed leverage varies so much by industry — a utility with stable, predictable cash flows and low distress costs can sustain far more debt than a volatile, R&D-intensive firm with high distress costs. These differences are a primary driver of the choice of leverage, not something irrelevant to it.

v. TRUE. This is essentially the definition of financial distress: the risk that a firm won't be able to meet its contractual debt obligations (interest and principal repayments) as they fall due.

vi. FALSE. The correct benchmark is **EBIT** (earnings before interest and tax), not dividend payments. The tax benefit of debt comes from shielding taxable income with interest deductions — once interest expense exceeds EBIT, there's no more taxable income left to shield, so additional debt provides no further marginal tax benefit. Dividend payments (which are themselves paid out of after-tax profits, and aren't tax-deductible at all) have no bearing on this threshold.

How the 10 Marks Are Likely Allocated (per the exam's own stated rule)

Tier	Marks	What separates this tier
10	10	Selects exactly i, ii, and v, with no incorrect statements included.
Partial	1–9	Selects some but not all of i, ii, v, with no incorrect statements included (partial credit scales with how many of the three correct statements are identified).
0	0	Any incorrect statement (iii, iv, or vi) is included in the selection, regardless of how many correct statements are also selected — per the question's explicit marking rule.

Question 1 Total: 63 Marks

Question 2

Working Capital & the Cash Conversion Cycle (37 Marks)

Part (a)

15 Marks

Westerly Industries has the following financial information. What is its cash conversion cycle?

Sales	100,000
Cost of Goods Sold	80,000
Accounts Receivable	30,000
Inventory	15,000
Accounts Payable	40,000

FULL MODEL ANSWER

The cash conversion cycle has three components, each measured in days:

CCC = Days Sales Outstanding (DSO) + Days Inventory Outstanding (DIO) − Days Payables Outstanding (DPO)

Component	Formula	Calculation	Days
DSO	AR ÷ Sales × 365	30,000 ÷ 100,000 × 365	109.50
DIO	Inventory ÷ COGS × 365	15,000 ÷ 80,000 × 365	68.44
DPO	AP ÷ COGS × 365	40,000 ÷ 80,000 × 365	182.50

CCC = 109.50 + 68.44 − 182.50

CCC ≈ −4.56 days

A **negative** cash conversion cycle means Westerly is, on average, paid by its suppliers' generosity rather than the reverse — it takes longer to pay its own suppliers (182.5 days) than the combined time it takes to sell inventory and collect from customers (109.5 + 68.4 = 177.9 days). In effect, suppliers are financing Westerly's operating cycle rather than Westerly needing to fund it itself, which is a genuinely favourable (if unusual) working capital position — commonly seen in retail-type businesses with strong supplier payment terms relative to how quickly they turn over stock and collect from customers.

How the 15 Marks Are Likely Allocated

Tier	Marks	What separates this tier
12–15	12–15	Correctly calculates all three components (DSO, DIO, DPO) using COGS as the base for DIO and DPO (not Sales), and correctly combines them to reach the negative CCC of ≈−4.56 days, with a correct interpretation of what the negative sign means.
7–11	7–11	Correct formulas but one component uses the wrong base (e.g. Sales instead of COGS for DIO or DPO), carried through under the own-figure rule.
3–6	3–6	Attempts the calculation but with the DPO term added rather than subtracted, or a fundamental misunderstanding of which figures belong in each ratio.
0–2	0–2	No attempt, or no understanding of the cash conversion cycle formula.

Part (b)

12 Marks

The Manana Corporation had sales of €69 million this year. Its accounts receivable balance averaged €7 million. How long, on average, does it take the firm to collect on its sales?

FULL MODEL ANSWER

This is asking for Days Sales Outstanding (DSO) — the average collection period:

DSO = Accounts Receivable ÷ Sales × 365
DSO = €7,000,000 ÷ €69,000,000 × 365
DSO ≈ 37.03 days

On average, Manana takes just over 37 days to convert a sale into actual cash in hand — useful context for assessing whether its credit terms and collection practices are keeping pace with the credit it's effectively extending to customers.

How the 12 Marks Are Likely Allocated

Tier	Marks	What separates this tier
10–12	10–12	Correctly applies the DSO formula and reaches ≈37.03 days.
5–9	5–9	Correct formula but a minor arithmetic or unit error (e.g. using 360 days instead of 365, which is an acceptable convention in some contexts and should still earn most of the marks if applied consistently).
0–4	0–4	No attempt, or the wrong ratio entirely (e.g. inventory or payables turnover used instead of receivables).

Part (c) **10 Marks**

Which one of the following statements is FALSE? Briefly explain why it is false.

- A firm's cash cycle is the length of time between when the firm pays cash to purchase its initial inventory and when it receives cash from the sale of the output produced from that inventory.
- The longer a firm's cash cycle, the more working capital it has, and the more cash it needs to carry to conduct its daily operations.
- Pharmaceutical companies tend not to hold much cash and securities.
- Any reduction in working capital requirements generates a positive free cash flow that the firm can distribute immediately to shareholders.

FULL MODEL ANSWER

Statement iii is FALSE.

Why iii is false: Pharmaceutical companies are, in reality, well known for holding **substantial** cash and marketable securities balances — not small ones. The reasoning is specific to the industry's risk profile: pharmaceutical firms face large, uncertain, front-loaded R&D spending with binary outcomes (a drug either clears clinical trials and regulatory approval or it doesn't), long development timelines, and volatile future cash flows once patents eventually expire. Holding large cash reserves gives these firms a buffer to keep funding research through setbacks without being forced to raise external capital on unfavourable terms at exactly the moment a trial fails or a funding round is most needed. Technology companies are frequently cited alongside pharmaceuticals as an industry that holds unusually large cash balances for the same underlying reason — high cash flow volatility and valuable, uncertain growth options that are worth preserving financial flexibility for.

Why the other three statements are true: Statement i is the standard definition of the cash cycle used in corporate finance texts. Statement ii follows directly from the definition: a longer gap between paying for inventory and collecting cash from its eventual sale means more capital is tied up in the operating cycle at any point in time, requiring more working capital funding. Statement iv reflects the direct mechanical relationship between working capital and free cash flow (free cash flow is calculated net of the change in net working capital, so a reduction in working capital requirements is, by construction, an addition to free cash flow, which the firm is then free to use for a distribution to shareholders if it chooses to).

How the 10 Marks Are Likely Allocated

Tier	Marks	What separates this tier
8–10	8–10	Correctly identifies statement iii as false, and explains that pharmaceutical (and similarly, technology) companies actually hold large cash and securities balances because of high R&D uncertainty and cash flow volatility, rather than small ones.
4–7	4–7	Correctly identifies statement iii as false but with a weaker or generic explanation not grounded in the industry-specific reasoning.
0–3	0–3	Identifies the wrong statement as false, or no attempt.

Question 2 Total: 37 Marks